IDENTIFICATION OF TRANSCRIPTOMIC BIOMARKERS RELATED TO ONCOGENIC PROCESSES ASSOCIATED WITH OROPHARYNGEAL CANCER IN HPV-POSITIVE PATIENTS.

LUISA FERNANDA CANTILLO (MASTER IN MICROBIOLOGY, UNAL), ANDRÉS GARCÍA LÓPEZ, LILIANA LÓPEZ KLEINE (DEPARTMENT OF STATISTICS, UNAL), JOSEFA A. RODRÍGUEZ (CANCER BIOLOGY RESEARCH GROUP, INC)

UNIVERSIDAD NACIONAL DE COLOMBIA

iSCB INTERNATIONAL SOCIETY FOR COMPUTATIONAL BIOLOGY

INTRODUCTION

Human papillomavirus (HPV) is one of the most common sexually transmitted infections and can lead to cancer, particularly oropharyngeal cancer (OPC), whose incidence has increased in recent decades. This study analyzed data from patients with head and neck squamous cell carcinoma (HNSCC) who are HPV-positive and HPV-negative to identify transcriptomic biomarkers. Five datasets from the NCBI GEO database were used, filtering samples and genes to obtain differentially expressed genes (DEGs). A co-expression network was constructed, revealing key interactions between genes and identifying several candidates for future validation. These findings could contribute to new approaches for the diagnosis and prognosis of HPV-associated HNSCC.

METHODOLOGY

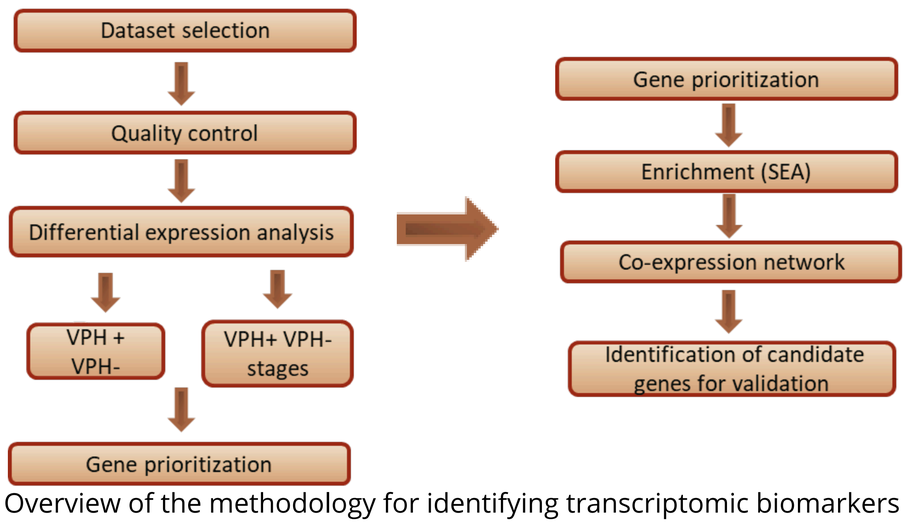


Figure 1. Overview of the methodology for identifying transcriptomic biomarkers related to oncogenic processes in HPV-positive oropharyngeal cancer.

Table 1. Analyzed datasets obtained from the Gene Expression Omnibus (GEO).

Dataset	Type	Number of Samples	Source	Citation
GSE72536	RNAseq	23 tumor samples from patients with HNSCC, of which 10 are HPV+ and 13 are HPV-	University of Southampton, UK, 2015	(Wood et al., 2016)
GSE205308	RNAseq	72 tumor samples, of which 20 HPV+ and 20 HPV- were selected	Helmholtz Center Munich, Germany, 2022	(Lombardo et al., 2022)
GE74927	RNAseq	36 tumor samples where 16 are HPV+ and 16 are HPV-	University of Michigan, USA, 2015	(Zhang et al., 2016)
GSE139324	scRNAseq	63 samples, excluding results from tumor samples; only 8 HPV+ and 8 HPV- peripheral blood samples will be analyzed	University of Pittsburgh, USA, 2019	(Cillo et al., 2020b)
GSE209670	miRNA	32 samples, selecting 20 samples from plasma and saliva, including 12 from HPV- HNSCC patients and 8 from HPV+ HNSCC patients	University of Ulm, Germany, 2022	(Hofmann et al., 2022)

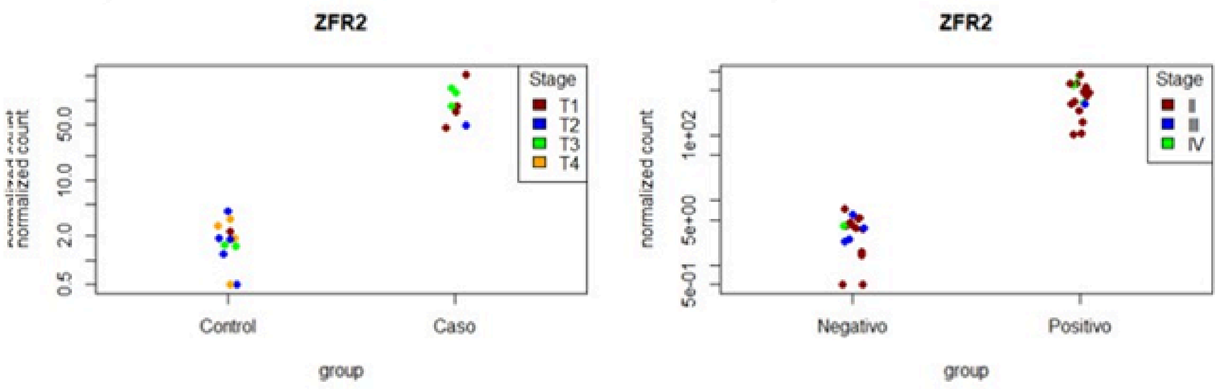
RESULTS AND DISCUSSION

A total of 2,081 differentially expressed genes (DEGs) and 6 miRNAs were identified (see Table 2). A preliminary enrichment analysis of the gene lists revealed significant associations with adhesion, carcinogenesis, the nervous system, and cellular transcription.

Table 2. Number of differentially expressed genes in each data set.

Dataset	GSE205308	GSE74927	GSE72536	GSE139324	GSE209670
Upregulated	50	787	246	0	NA
Downregulated	33	1008	257	0	NA
Total DEGs	83	1795	203	0	6

Using disease stage as a covariate in the differential expression analysis, we identified 1,103 DEGs and 1 miRNA. Figure 2 shows that no distinct groupings emerged, indicating that gene expression differences are not linked to the disease stage.



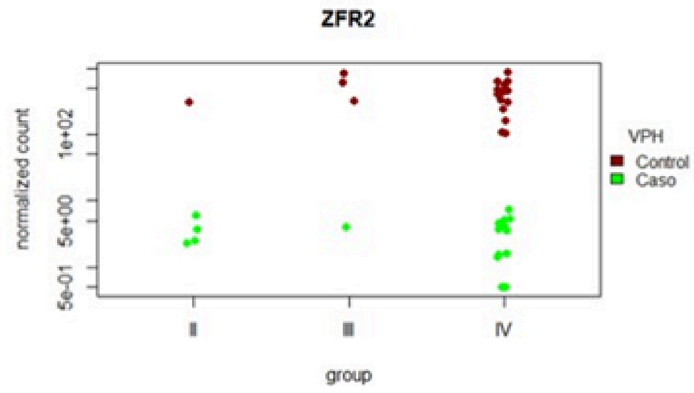


Figure 2. Gene expression and presence of HPV in different stages of oropharyngeal cancer. The figure displays gene expression levels, with points indicating HPV presence and colors representing disease stages. These results suggest that gene expression differences are more closely linked to HPV status than to the disease stages.

A new network was constructed by incorporating the miRNAs into the adjacency matrix, allowing for the visualization of a bipartite network of genes and miRNAs. This integration enabled us to explore whether the miRNAs regulate the expression of selected genes. The resulting network reveals that the miRNAs regulate several of these genes, indicating a functional relationship that may be implicated in HPV-mediated carcinogenesis. Additionally, some topological measures were calculated, highlighting the connectivity and hub genes within the network.

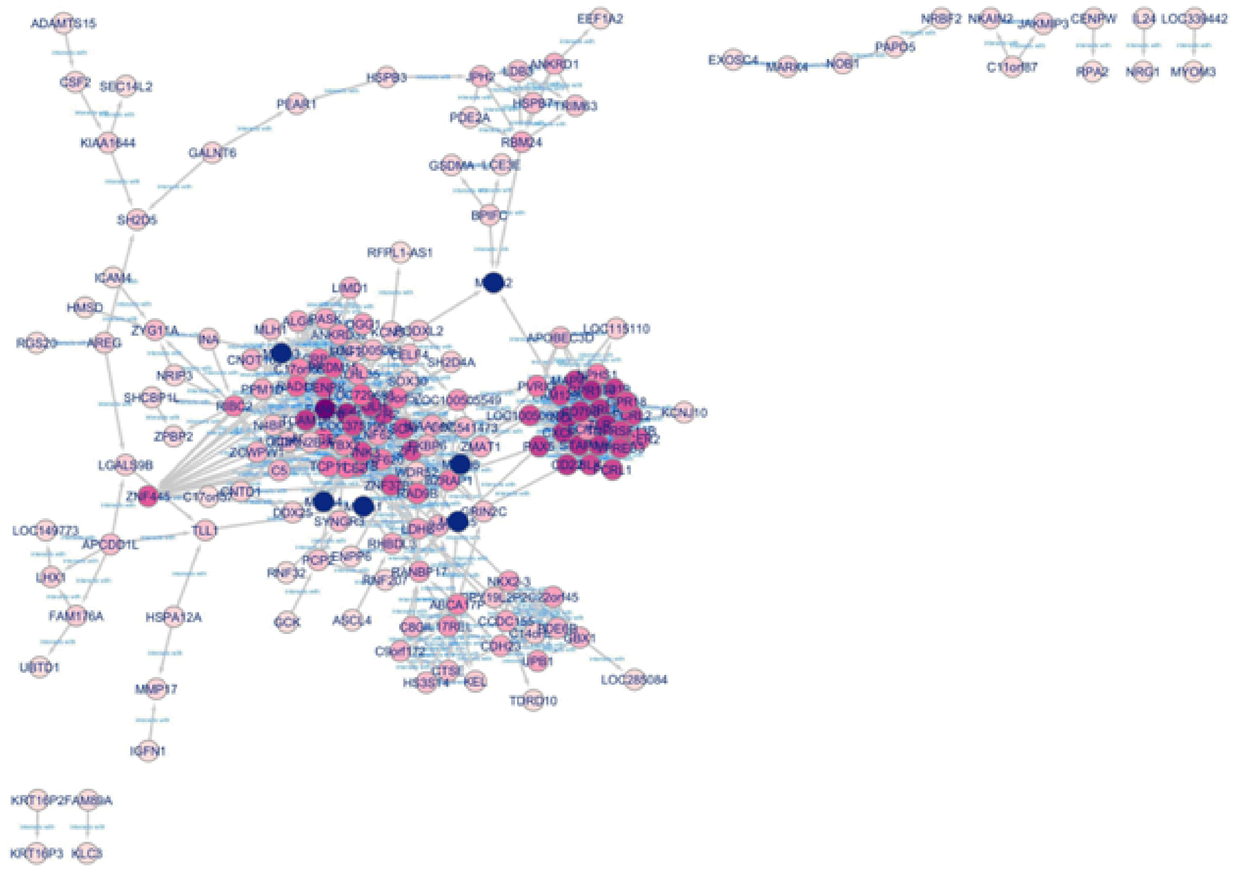


Figure 3. Co-expression network of differentially expressed genes and miRNAs in HPV+ HNSCC patients. The figure illustrates the co-expression network modules, with miRNAs shown in blue and genes in pink. Nodes are colored in varying shades of pink based on their degree in the network, with darker shades indicating higher connectivity. The six differentially expressed miRNAs are connected to genes within the network, suggesting their potential regulatory role in gene expression

FUTURE WORK

Five candidate genes were identified for validation in databases and future experimental confirmation through RT-qPCR. These genes may offer new insights into the molecular mechanisms underlying HPV-associated HNSCC and could potentially serve as biomarkers for the diagnosis and prognosis of this disease.

CONCLUSIONS

The gene expression analysis showed that differences in gene expression are more closely related to the presence of HPV than to the stages of oropharyngeal cancer.

The integration of miRNAs into the co-expression network revealed that dysregulated genes and miRNAs act as transcriptional regulators, emphasizing HPV's role in genetic modulation. Topological measures further supported the prioritization of these genes, enabling the identification of central genes that could serve as potential biomarkers for the pathogenesis of HPV-related oropharyngeal cancer.